



Database Management Systems

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Data Models

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Data Models



- ❑ A **framework** that defines how data is stored, organized and manipulated in a database. *son*
- ❑ In other word its a collection of conceptual tools for describing data, data relationships, data semantics, and consistency constraints. *sc*
- ❑ **High Level Data Models:** Entity-Relationship (E-R) Model, Unified Modeling Language (UML)
- ❑ **Representational Data Models:** Network, Hierarchical, Relational.
- ❑ **Physical Data Models:** Data structures

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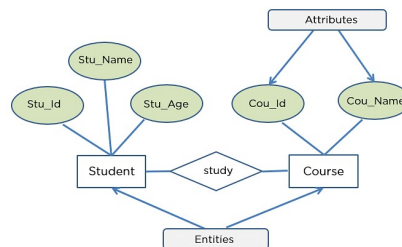
Entity-Relationship Model (ER Model)



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Entity-Relationship (ER) Model

- ❑ It provides a high level, graphical view of data and its structure.
- ❑ **Simplifies Database Design:** Easy to visualize data structure and relationships.
- ❑ **Avoids Redundancy:** Proper relationships minimize duplicate data.
- ❑ **Foundation for Database Implementation:** Translates directly into a relational schema.

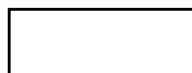


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Key Component of ER Model

❑ Entity:

- **Entity:** An entity is a '*thing*' or '*object*' in the real world that is distinguishable from all other objects. It is represented by *set of attributes*.
- **Entity Set:** Set of entities of the same type that share the same **properties**, or attributes.
 - **Strong Entities:** Exist independently (e.g. employee)
 - **Weak Entities:** Depend on strong entity (e.g. Dependent of employee).



Entity



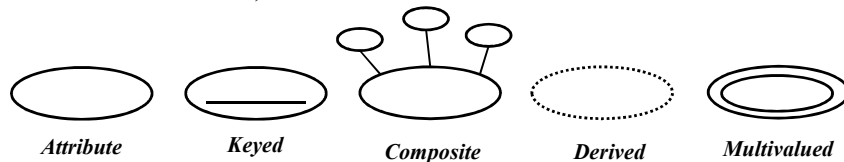
Weak Entity

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Key Component of ER Model

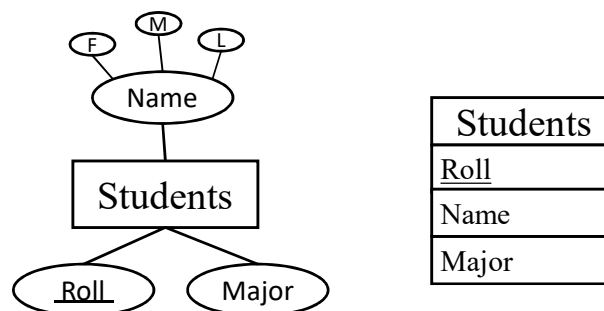
Attribute: Characteristics of entities.

- **Simple:** Atomic values (e.g. Name, Age)
- **Keyed:** an attribute which can uniquely identify an entity (e.g. Roll number).
- **Composite:** Made up of multiple simple attributes (e.g. Full Name = First Name + Surname).
- **Derived:** Calculated from other attributes (e.g. Age can be derived from DoB).
- **Multivalued:** can have multiple values (e.g. Mobile Number).



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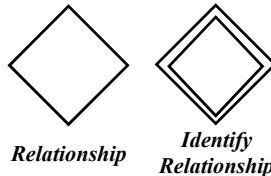
Example: Entity and Attribute



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Key Component of ER Model

- ❑ **Relationship:** Represents **association** between entities
- ❑ **Cardinalities:** Number of entities that can participate in a relationship.
 - **One-to-One (1:1):** A person having single passport.
 - **One-to-Many (1:N):** A department having many employees.
 - **Many-to-Many (M:N):** Students enrolled in courses
 - **Many-to-One (N:1):** Many employees associated to single department.

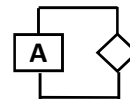
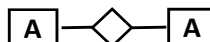


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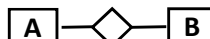
Degree of Relationship

- ❑ **Degree of relationship:** How many entity sets participated in a relationship:

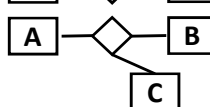
❑ **Unary:** Only one entity.



❑ **Binary:** Two entity sets.



❑ **Ternary:** Three entities.



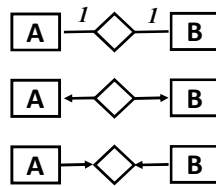
- ❑ **Min = 1, Max = N** (difficult to manage when converted in relational model).

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One-to-One Relationship: Example



- ❑ **Rule 1:** An entity in set A can be associated with at most one entity in set B.
- ❑ **Rule 2:** An entity in set B can be associated with at most one entity in set A.

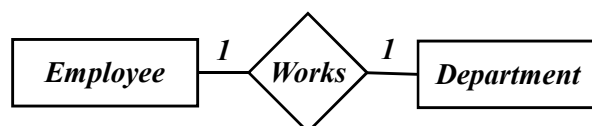


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One-to-One Relationship: Example



- ❑ **Example:** Employee and Department



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One-to-One Relationship: Example



- ❑ What will be attributes of *Employee* and *Department*?



EID	Name	Age
E1	DNY	31
E2	DPM	50
E3	SPY	45
E4	BDS	50

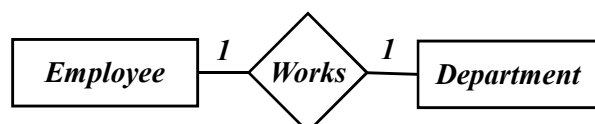
DID	Name	LOC
D1	IT	31
D2	ADMIN	50
D3	HR	45

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One-to-One Relationship: Example



- ❑ What will be primary key for this relations?



<u>EID</u>	Name	Age
E1	DNY	31
E2	DPM	50
E3	SPY	45
E4	BDS	50

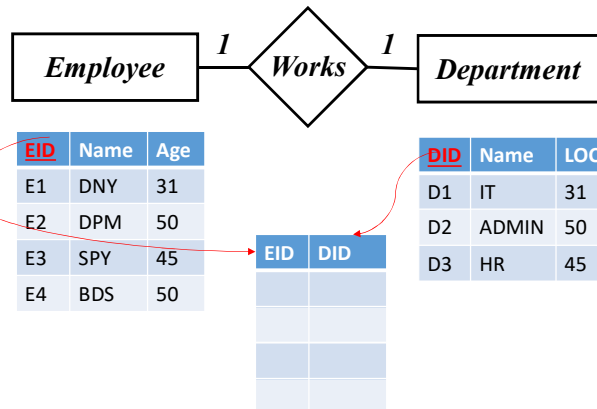
<u>DID</u>	Name	LOC
D1	IT	31
D2	ADMIN	50
D3	HR	45

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One-to-One Relationship: Example



❑ Attributes of works table/relation?

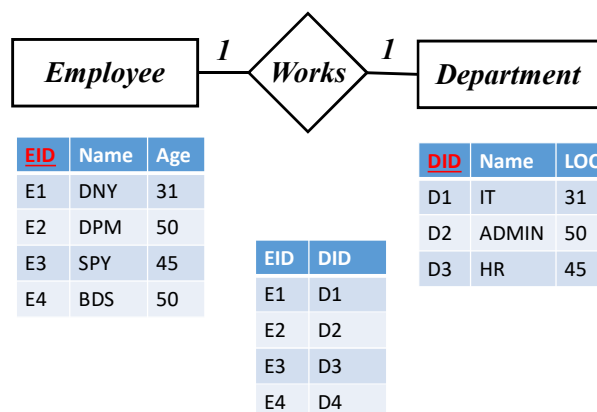


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One-to-One Relationship: Example



❑ Records for works?

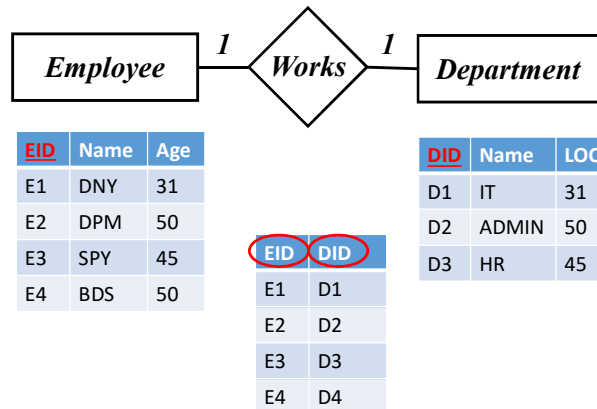


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One-to-One Relationship: Example



- ☐ Primary key for works relations?

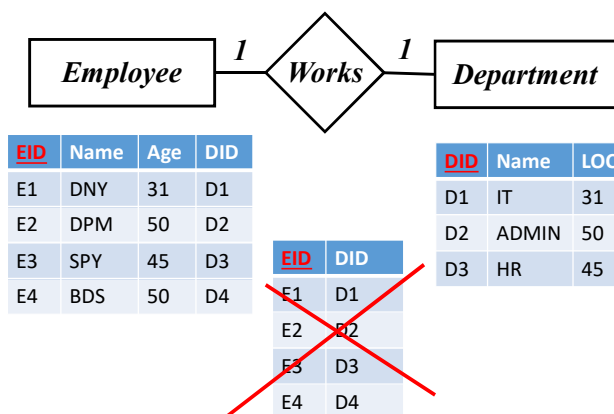


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One-to-One Relationship: Example



- ☐ Can we reduce the number of relations?

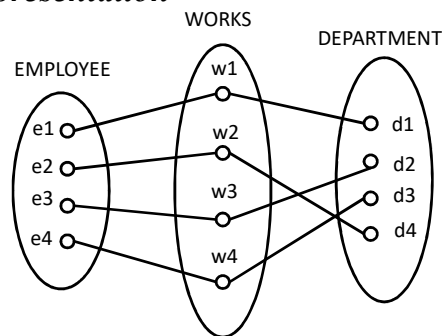


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One-to-One Relationship: Example



Simple Representation



Possible Questions:

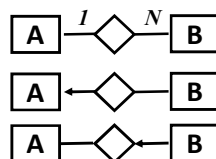
- What will be primary key if relationship is one-to-one?
- Can we reduce the table?

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One-to-Many Cardinality: Example



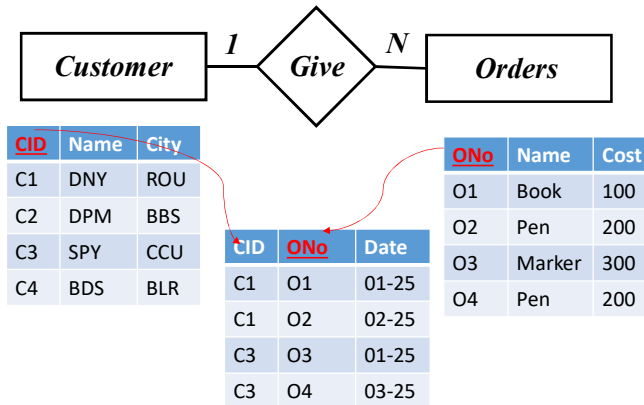
- **Rule 1:** An entity in set A can be associated with any number (zero or more) of entities in set B.
- **Rule 2:** An entity in set B can be associated with at most one entity in set A.



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One-to-Many Relationship: Example

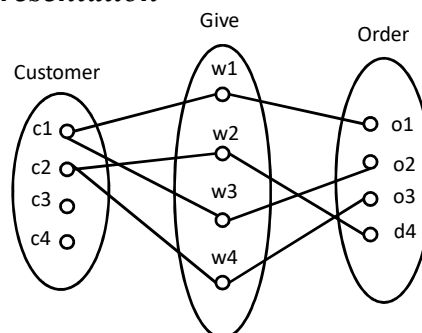
- ❑ Customer and Order: A customer can give many orders but the order can be given by single customer only.



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One-to-Many Relationship: Example

- ❑ Simple Representation



- ❑ Possible Questions:

- What will be primary key if relationship is one-to-many?
- Can we reduce the table?

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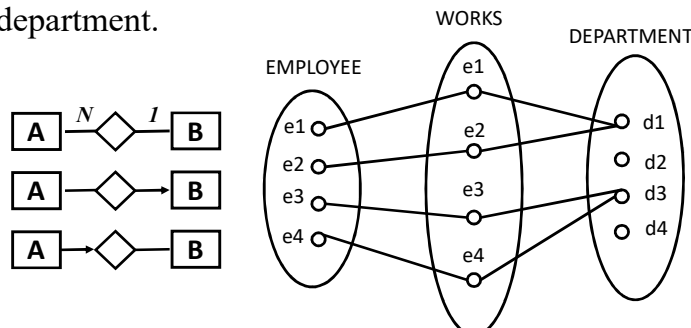
One-to-Many Relationship: Example

- ❑ Primary key in *Give*?
 - ❑ **OID**: (always choose the primary key of many side)
- ❑ Can we reduce the relations?
 - ❑ **Yes**: Add the Give table entries to Order table.
 - ❑ **We can not merge Customer and Give table.**
- ❑ **Note**: Just opposite to this will be many-to-One relationship.

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Many-to-One Cardinality: Example

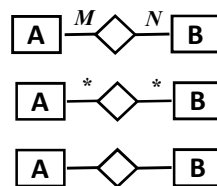
- ❑ **Rule 1**: An entity in set A can be associated with at most one entity in set B.
- ❑ **Rule 2**: An entity in set B can be associated with any number (zero or more) of entities in set A.
- ❑ **Example**: Many employees can work for a single department.



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Many-to-Many Cardinality: Example

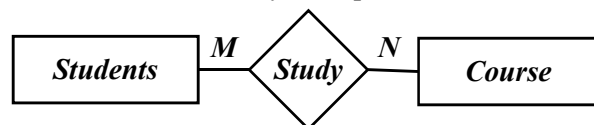
- ❑ **Rule 1:** An entity in set A can be associated with any number (zero or more) of entities in set B.
- ❑ **Rule 2:** An entity in set B can be associated with any number (zero or more) of entities in set A.



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Many-to-Many Relationship: Example

- ❑ Student - Course:
 - A student can study multiple courses.
 - A course can be studied by multiple students.



SID	Name	Age
S1	DNY	19
S2	DPM	20
S3	SPY	21
S4	BDS	22

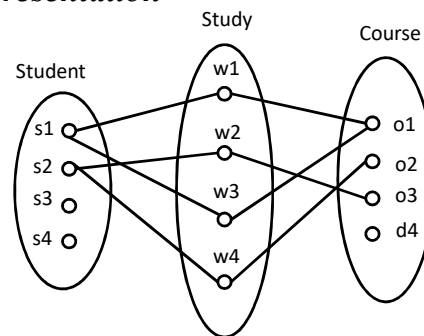
SID	ONo
S1	C1
S1	C2
S2	C3
S2	C1

CID	Name	Credit
C1	Math	3
C2	Phy	3
C3	DB	4
C4	C++	4

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Many-to-Many Relationship: Example

Simple Representation



Possible Questions:

- What will be primary key if relationship is many-to-many?
- Can we reduce the table?

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Many-to-Many Relationship: Example

Primary key in *Study*?

☐ *Composite: (SID+CID)*

Can we reduce the relations?

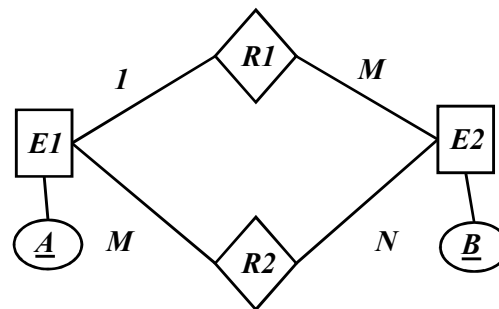
☐ *No reduction of table.*

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E-R Model: GATE Question (2005)



- ☐ Min number of tables requires to represent this E-R model into relational table?

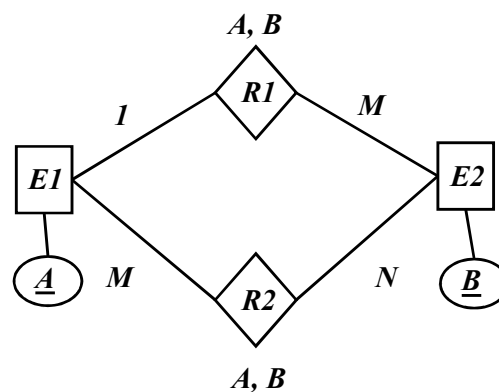


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E-R Model: GATE Question (2005)



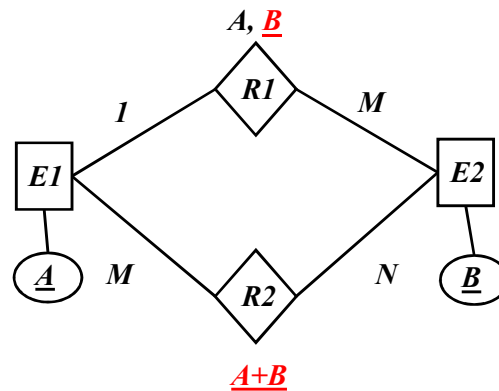
- ☐ Attributes in R1?
☐ Attributes in R2?



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E-R Model: GATE Question (2005)

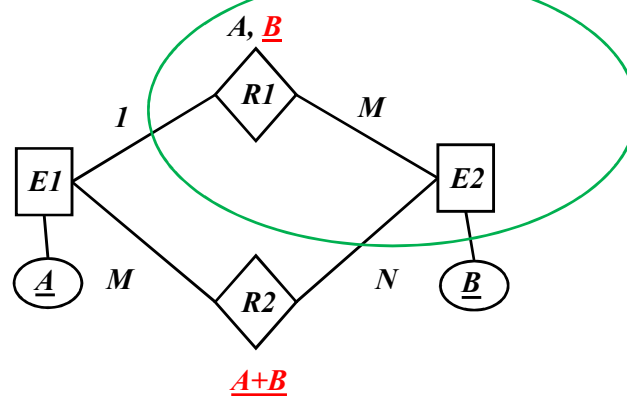
- ☐ Primary key in R1?
- ☐ Primary key in R2?



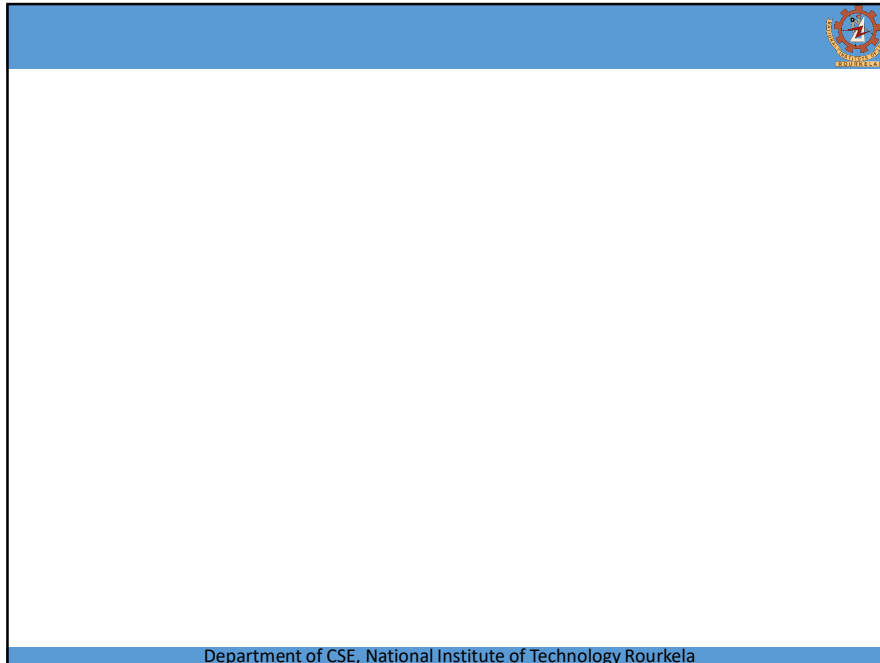
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E-R Model: GATE Question (2005)

- ☐ Which relations can be combined?
- ☐ Total number of tables?



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Structural Constraints

- ☐ **Structural constraints** define the rules and restrictions for how entities participate in relationships.
- ☐ These constraints help ensure data integrity in the database design.
- ☐ The two primary structural constraints are:
 - **Cardinality Constraints:** Define the number of entity set (E) that can participate in a relationship (R).
 - **Participation Constraints / Modality:** Define whether the existence of an entity depends on its participation in the relationship.

Participation Constraints



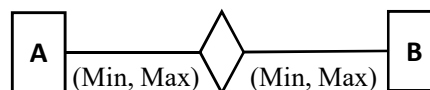
- ❑ **Participation Constraints:** : define whether the existence of an entity depends on its participation in the relationship.
- It defines *Maximum* and *Minimum* number of relationship instance that each entity *can/must* participate.
 - **Examples:**
 - A employee can work at most one department.
 - A department must have at least 10 employees but not more than 20.
 - A faculty can have min 2 PhD and Maximum 5 PhD.
 - A student must enroll for at least 3 courses but not more than 6 courses.

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Participation Constraints



- ❑ **Maximum Cardinality:** It defines the maximum number of time an entity participating in a relationship.
- ❑ **Minimum Cardinality:** It defines the minimum number of time an entity participating in a relationship.

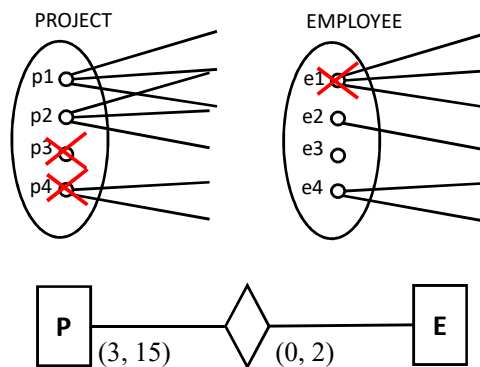


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Participation Constraints

❑ **Example:** Project and Employee

- Project: Minimum 3 employee, Maximum 15 Employee.
- Employee: Minimum 0 project, Maximum 2 projects.

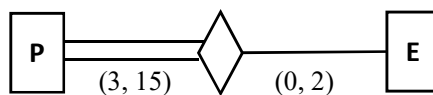


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Participation Constraints

❑ **Total Participation:** Every instance of an entity must participate in the relationship.

- **Notation:** Represented by a double line connecting the entity to the relationship.
- **Example:** Every project have at three employee.
- **Note:** Minimum participation is one.

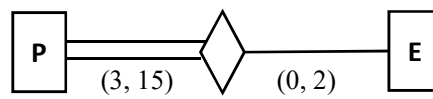


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Participation Constraints

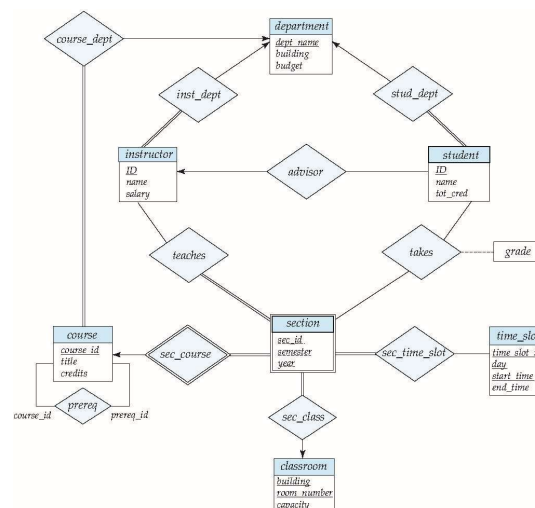
❑ **Partial Participation:** Some instances of the entity may not participate in the relationship.

- **Notation:** Represented by a single line.
- **Example:** Every project have at three employee.
- **Note:** Minimum participation is zero.



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ER Diagram: Example



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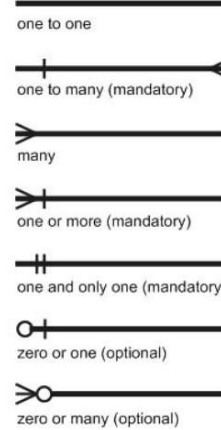
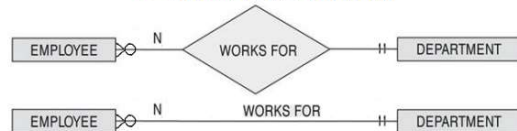
Work out an example for this relations



Cardinality Ratio representation: cardinality of two entities written in distance from them (looks like swapped)!



Min-Max Representation: Provides more information! Possible to draw for n -ary relationship.



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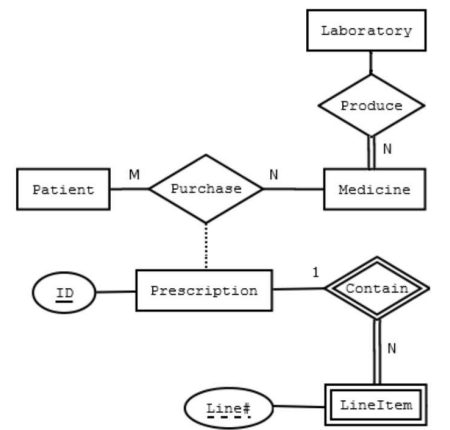
Work out an Example for this relations

OTHER RELATIONSHIP CARDINALITY NOTATION

Notation	Zero or One Relationship	One and Only One	Zero or Many Relationship	One or Many Relationship
Crow's Foot Notation				
Arrow Notation				
Bachman Notation				
ADW				
Oracle				

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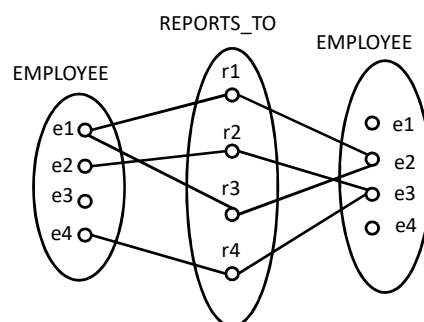
Explain this ER Diagram?



- Medicine participates totally in the Produce relationship, meaning that a medicine can't exist unless Produced by a laboratory. In contrast, a laboratory can exist without producing any medicine - Laboratory participates partially in the Produce relationship.
- a Patient can Purchase Medicine with or without a Prescription. A Purchase can't exist without a Patient and Medicine, but a Prescription is optional in Purchase relationship.
- a Prescription contains LineItems which are identified by the Prescription's key (ID) and a line number (Line#) together, and not by Line# only. In other words, the LineItems will have a composite key (ID, Line#).

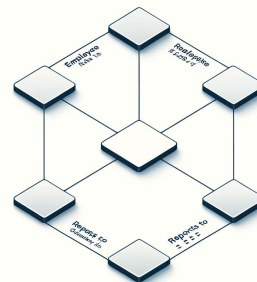
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Recursive Relationship



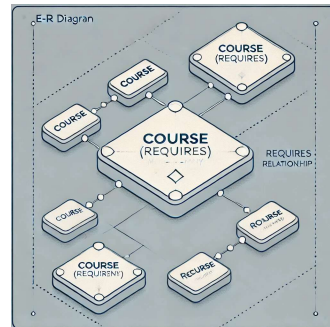
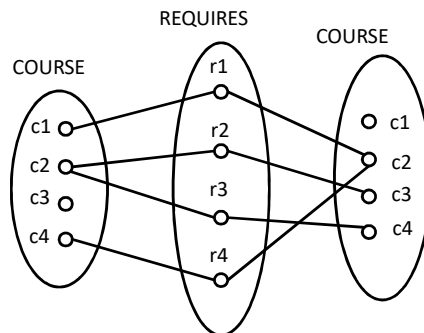
Organization Rule:

- An employee may report to another employee (a manager).
- A manager can have multiple subordinates, but each employee has only one direct manager.



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Recursive Relationship



Organization Rule:

- A course may have another course as its prerequisite.
- A course can require multiple other courses, and the same course can be a prerequisite for multiple other courses.

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Entity Sets (GATE 2020)

Q. Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?

- Diamonds with double/bold border
- Rectangles with double/bold border
- Ovals with double/bold border
- Ovals that contain underlined identifiers

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Entity Sets (GATE 2020)

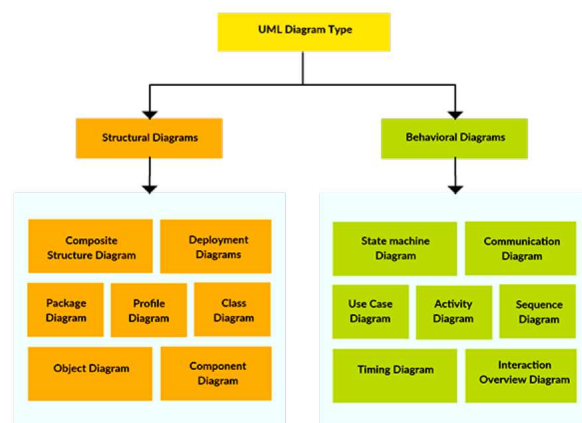
Q. Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?

- (a) **Diamonds with double/bold border**
- (b) Rectangles with double/bold border
- (c) Ovals with double/bold border
- (d) Ovals that contain underlined identifiers

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UML (Unified Modeling Language)

□ A standardized modeling language used to visualize, specify, construct, and document the structure and behavior of a system.



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UML (Unified Modeling Language)



□ Common Use:

- **System Design:** To design software systems and ensure all components work together effectively.
- **Documentation:** For creating detailed documentation for existing systems.
- **Communication:** To convey ideas between developers, stakeholders, and clients.
- **Analysis:** To analyze existing systems or explore new ones.

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UML (Unified Modeling Language)



□ Structural Diagram:

- **Class Diagram:** Represents the static structure of a system, showing classes, their attributes, methods, and relationships.
- **Object Diagram:** Depicts a snapshot of objects and their relationships at a particular point in time.
- **Component Diagram:** Shows the organization and dependencies of software components.
- **Deployment Diagram:** Illustrates the physical deployment of artifacts on nodes (e.g., servers, devices).
- **Package Diagram:** Groups related elements into packages to organize complex systems.
- **Composite Structure Diagram:** Represents the internal structure of a class and collaborations.

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UML (Unified Modeling Language)



□ Behavioral Diagram:

- ***Use Case Diagram:*** Visualizes functional requirements, actors, and their interactions with the system.
- ***Sequence Diagram:*** Details the interaction between objects in a sequential order.
- ***Activity Diagram:*** Models workflows, processes, and business logic.
- ***State Machine Diagram:*** Describes the states and transitions of an object during its lifecycle.
- ***Communication Diagram:*** Focuses on the interactions between objects in terms of messages.

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Relational Model



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Relational Model



- ❑ The **Relational Model** is a fundamental framework for organizing data in databases.
- ❑ Introduced by **Edgar F. Codd in 1970**.
- ❑ It uses a **mathematical approach** to represent data and relationships in the form of tables (relation).

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Key Concept of Relational Model



- ❑ **Relation Schema:**
 - Defines the structure of a relation, including its name, attributes, and domains.
 - Students (ID: INT, Name: VARCHAR, Major: VARCHAR)
 - Enrollments(SN: INT, ID: INT, CourseID: VARCHAR)

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE
103	Surya	MTH
104	Ashish	MTH
105	Anshu	PHY
106	Sumit	CHY

SN	ID	CourseID
1	101	CS101
2	102	CS102
3	103	MT101
4	102	EE101
5	104	CH102
6	105	CH101

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Key Concept of Relational Model



❑ Relation (Table):

- A relation is a two-dimensional table consisting of rows (tuples) and columns (attributes).
- Each table represents an entity or a relationship.
- Each table is assigned a unique name.

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE
103	Surya	MTH
104	Ashish	MTH
105	Anshu	PHY
106	Sumit	CHY

SN	ID	CourseID
1	101	CS101
2	102	CS102
3	103	MT101
4	102	EE101
5	104	CH102
6	105	CH101

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Key Concept of Relational Model



❑ Tuple (Row):

- A single row in a table represents a unique instance of the entity.
- (1, Vineet, CSE)

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE
103	Surya	MTH
104	Ashish	MTH
105	Anshu	PHY
106	Sumit	CHY

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Key Concept of Relational Model



❑ *Attribute (Column):*

- Each column in a table represents a property or characteristic of the entity.
- ID, Name, Major

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE
103	Surya	MTH
104	Ashish	MTH
105	Anshu	PHY
106	Sumit	CHY

SN	ID	CourseID
1	101	CS101
2	102	CS102
3	103	MT101
4	102	EE101
5	104	CH102
6	105	CH101

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Key Concept of Relational Model



❑ *Domain:*

- The set of allowable values for an attribute
- Example: Domain of ID = {101, 102, 103, ...}

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE
103	Surya	MTH
104	Ashish	MTH
105	Anshu	PHY
106	Sumit	CHY

SN	ID	CourseID
1	101	CS101
2	102	CS102
3	103	MT101
4	102	EE101
5	104	CH102
6	105	CH101

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Key Concept of Relational Model



❑ Primary Key:

- A column (or a combination of columns) that uniquely identifies each row in the table.
- Example: ID in relation Students, SN in relation Enrollments.

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE
103	Surya	MTH
104	Ashish	MTH
105	Anshu	PHY
106	Sumit	CHY

SN	ID	CourseID
1	101	CS101
2	102	CS102
3	103	MT101
4	102	EE101
5	104	CH102
6	105	CH101

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Key Concept of Relational Model



❑ Foreign Key:

- A column that establishes a relationship between two tables by referencing the primary key of another table.
- Example: ID

ID	Name	Major
101	Vineet	CSE
102	Ayush	CSE
103	Surya	MTH
104	Ashish	MTH
105	Anshu	PHY
106	Sumit	CHY

SN	ID	CourseID
1	101	CS101
2	102	CS102
3	103	MT101
4	102	EE101
5	104	CH102
6	105	CH101

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Key Concept of Relational Model



❑ **Integrity Constraints:** Rules to ensure the accuracy and consistency of data.

- **Domain Constraint:** Attributes must contain values from the specified domain.
- **Key Constraint:** Every row must have a unique primary key.
- **Referential Integrity:** Foreign keys must reference existing primary keys.

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ER Diagram to Relational Model



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ER Model vs Relational Model



ER Model	Relational Model
Entity Type	Entity Relation
1:1 or 1:N relationship type	foreign key (or relation)
M:N relationship type	a relation with 2 foreign keys
n-ary relationship type	a relation with n foreign keys
simple attribute	attribute in relation
composite attribute	set of attributes in relation
multivalued attribute	relation and foreign key
value set	domain
key attributes	keys / candidate keys / primary keys

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ER Model vs Relational Model



□ Entity Type:

- ER Model → collection of similar entities (i.e. Student, Course).
- Relational Model → Tables or Relations.

□ 1:1 or 1:N Relationship Type:

- ER Model → define relation between entities one entity is related to at most one (1:1) or multiple entities (1:N).
- Relational Model → implemented using foreign keys or as relations to establish the connection.

□ M:N Relationship Type:

- ER Model → link multiple entities in both directions.
- Relational Model → implemented as a separate relation (table) that includes two foreign keys; one for each entity.

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ER Model vs Relational Model



❑ *N-ary Relationship Type:*

- ER Model → describes a relationship involving more than two entities (e.g., ternary relationships).
- Relational Model → a table with n foreign keys, each referencing the related entities.

❑ *Simple Attribute:*

- ER Model → contains atomic values (e.g., "Name" or "Age").
- Relational Model → Mapped directly to attributes in a table.

❑ *Composite Attribute:*

- ER Model → can be broken down into smaller components.
- Relational Model → represented as a set of attributes (multiple columns) in a relation.

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ER Model vs Relational Model



❑ *Multivalued Attribute:*

- ER Model → attribute can have multiple values
- Relational Model → multivalued attributes are implemented as a separate table (relation) with a foreign key.

❑ *Value Set:*

- ER Model → set of all possible values an attribute can take.
- Relational Model → referred to as the domain of the attribute.

❑ *Key Attributes:*

- ER Model → key attributes uniquely identify an entity within an entity set.
- Relational Model → these correspond to keys, which can include candidate keys and primary keys.

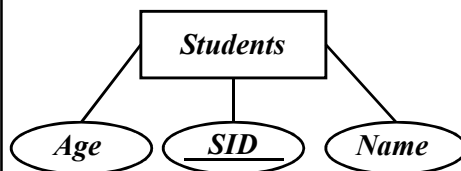
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ER Diagram → Relational Model



❑ Mapping Entity Sets to Tables:

- **Simple Entity:** Each entity becomes a table, with its attributes as columns.
 - Example: Student(Entity) → Student(Table) with columns SID, Name, Age.
- **Primary Key:** Use the key attribute of the entity as the primary key for the table.



<u>SID</u>	Name	Age
S1	Vineet	19
S2	Ayush	20
S3	Surya	21
S4	Ashish	19
S5	Anshu	20
S6	Sumit	22

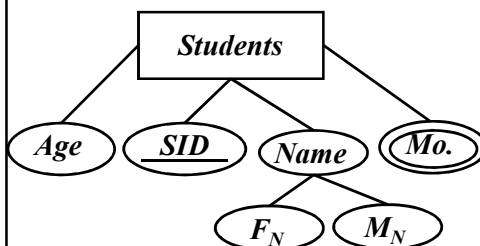
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ER Diagram → Relational Model



❑ Mapping Attributes:

- **Simple:** Directly becomes a column in the table.
- **Composite:** Split into individual columns.
- **Multivalued:** Create a new table including foreign key.



<u>SID</u>	F _N	M _N	Age
S1	Vin	eet	19
S2	Ay	ush	20
S3	Su	rya	21
S4	As	hish	19
S5	An	shu	20
S6	Su	nit	22

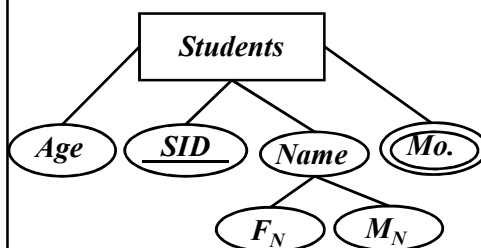
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ER Diagram → Relational Model



□ Mapping Attributes:

- **Simple:** Directly becomes a column in the table.
- **Composite:** Split into individual columns.
- **Multivalued:** Create a new table including foreign key.



SID	Mo.
S1	999
S1	998
S2	995
S3	994
S4	992
S4	991
S5	789
S6	885

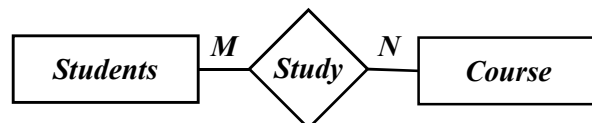
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ER Diagram → Relational Model



□ Mapping Relationship:

- **M:N** (Students can enroll in multiple courses, and a course can have multiple students)
- Create a separate table to represent the relationship.



SID	Name	Age
S1	Vineet	19
	Ayush	
S2	Surya	20
S3	Ashish	21
S4	Anshu	22

SID	CID	GD
S1	C1	A
S1	C2	B
S2	C3	B
S2	C1	A

CID	Name	Credit
C1	MTH	3
C2	PHY	3
C3	DB	4
C4	C++	4

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ER Diagram → Relational Model



Mapping Relationship:

- **1:N** (An instructor teaches multiple courses, but each course is taught by only one instructor).
- Create a separate table to represent the relationship.



<u>FID</u>	Name
F1	DNY
F2	SPY
F3	PDM
F4	BDS

<u>FID</u>	<u>CID</u>
F1	C1
F1	C3
F2	C4
F3	C2

<u>CID</u>	Name
C1	MTH
C2	PHY
C3	DB
C4	C++

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ER Diagram → Relational Model



Mapping Relationship:

- **1:N** (An instructor teaches multiple courses, but each course is taught by only one instructor).
- Create a separate table to represent the relationship.



<u>FID</u>	Name
F1	DNY
F2	SPY
F3	DPM
F4	BDS

<u>FID</u>	<u>CID</u>
F1	C1
F1	C3
F2	C4
F3	C2

<u>CID</u>	Name	<u>FID</u>
C1	MTH	F1
C2	PHY	F3
C3	DB	F1
C4	C++	F2

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ER Diagram → Relational Model



❑ Mapping Relationship:

- $N:1$ (Just opposite to $1:N$).
- $1:1$ (As discussed in previous lecture).

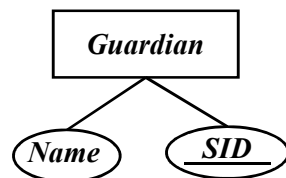
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ER Diagram → Relational Model



❑ Mapping Weak Entities:

- Guardians depend on Student. Each guardian is identified by the Student they are associated with and the name of the Guardian.
- Create a table for the weak entity and include a foreign key to the Student table.



<u>SID</u>	<u>G.Name</u>
S1	Nakara
S2	Nanda
S3	Nayak
S4	Pal
S5	Pal
S6	Nayak

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ER Diagram → Relational Model

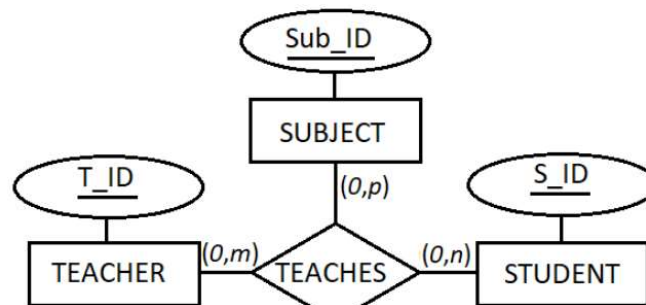


□ Mapping Constraints:

- **Primary Keys:** ensure uniqueness.
- **Foreign Keys:** maintain relationship.
- **Total Participation:** if every course must have an instructor, set FID in the Course table as NOT NULL.
- **Partial Participation:** If some students may not have guardians, Guardian is optional.

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ER Diagram → Relational Model



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Advantage of Relational Model



- ❑ **Simplicity:**
 - Data is represented in a tabular format, making it intuitive and easy to use.
- ❑ **Flexibility:**
 - Easy to add or modify tables, attributes, or relationships.
- ❑ **Data Independence:**
 - Logical and physical data independence ensures that changes in the schema do not affect application programs.
- ❑ **Mathematical Foundation:**
 - Based on relational algebra, allowing robust and precise query mechanisms.
- ❑ **Data Integrity:**
 - Enforces constraints to maintain data accuracy and consistency.

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Handwritten blue scribbles.

Disadvantage of Relational Model



- ❑ **Performance Issues:**
 - Complex queries can be slower compared to NoSQL models for large-scale or unstructured data.
- ❑ **Scalability:**
 - Horizontal scaling (adding servers) can be challenging.
- ❑ **Not Suitable for Complex Data Types:**
 - Handling multimedia, hierarchical, or graph-based data is cumbersome.

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GATE Questions

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GATE Questions

- ☐ ***Given the basic ER and relational models, which of the following is INCORRECT?***
 - (a) An attribute of an entity can have more than one value
 - (b) An attribute of an entity can be composite
 - (c) In a row of a relational table, an attribute can have more than one value
 - (d) In a row of a relational table, an attribute can have exactly one value or a NULL value
- ☐ ***[GATE 2012]***

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GATE Questions



❑ Given the basic ER and relational models, which of the following is **INCORRECT**?

- (a) An attribute of an entity can have more than one value
- (b) An attribute of an entity can be composite
- (c) In a row of a relational table, an attribute can have more than one value**
- (d) In a row of a relational table, an attribute can have exactly one value or a NULL value

❑ [GATE 2012]

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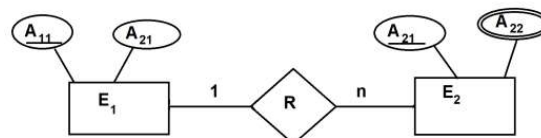
GATE Questions



❑ Q1. Maximum how many relations (tables) would be required when this ER diagram will be converted to relational model?

❑ Q2. Minimum how many relations (tables) would be required when this ER diagram will be converted to relational model?

❑ [GATE 2012]



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GATE Questions



❑ Q1. Maximum how many relations (tables) would be required when this ER diagram will be converted to relational model?

❑ 3

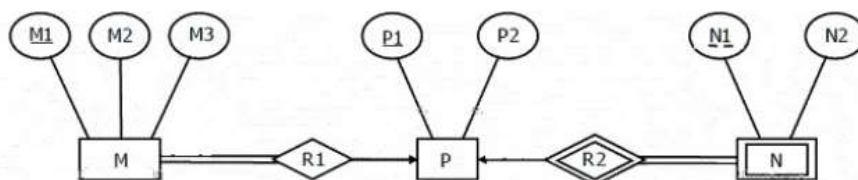
❑ Q2. Minimum how many relations (tables) would be required when this ER diagram will be converted to relational model?

❑ 2

❑ [GATE 2012]

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GATE Questions

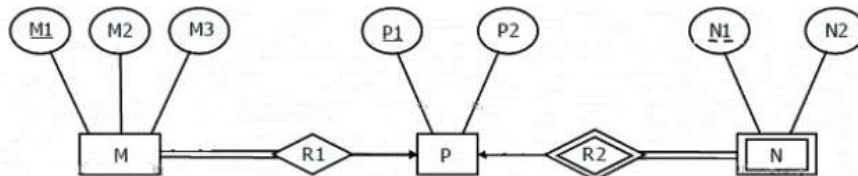


❑ Q1. Minimum how many relations (tables) would be required when this ER diagram will be converted to relational model?

❑ [GATE 2008]

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GATE Questions



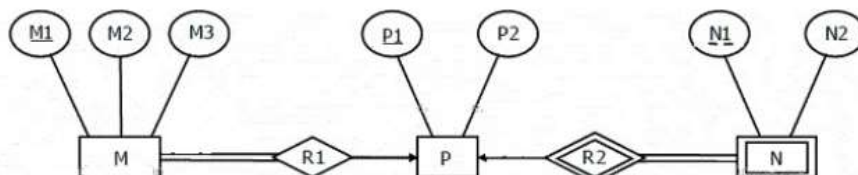
❑ Q1. Minimum how many relations (tables) would be required when this ER diagram will be converted to relational model?

❑ 2

❑ [GATE 2008]

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GATE Questions



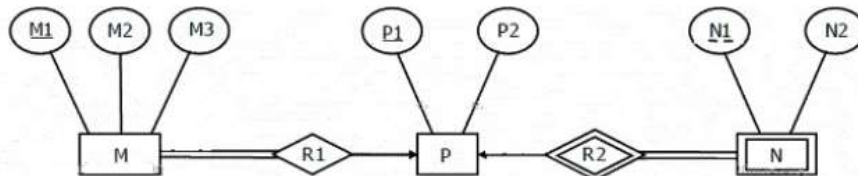
Q2. Which of the following is a correct attribute set for one of the tables for the correct answer to the above question?

- (a) M1, M2, M3, P1 (b) M1, P1, N1, N2
(c) M1, P1, N1 (d) M1, P1

❑ [GATE 2008]

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GATE Questions



Q2. Which of the following is a correct attribute set for one of the tables for the correct answer to the above question?

- (a) **M1, M2, M3, P1** (b) M1, P1, N1, N2
 (c) M1, P1, N1 (d) M1, P1

□ [GATE 2008]

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GATE Questions



□ **Q1.** E1 and E2 are two entities (with only simple attributes) having R1 and R2 as two relationships between them. R1 is a one-to-many and R2 is a many-to-many relationship. R1 and R2 do not have any attribute of their own. What is the minimum number of tables required to convert this ER model to relational model?

□ [GATE 2005]

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GATE Questions



❑ *Q1. E1 and E2 are two entities (with only simple attributes) having R1 and R2 as two relationships between them. R1 is a one-to-many and R2 is a many-to-many relationship. R1 and R2 do not have any attribute of their own. What is the minimum number of tables required to convert this ER model to relational model?*

❑ **3**

❑ *[GATE 2005]*

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GATE Questions



❑ *The table has two attributes A and C where A is the primary key and C is a foreign key referencing A with on-delete cascade. The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple (2, 4) is deleted is:*

❑ (a) (3,4),(6,4)

❑ (b) (5,2),(7,2)

❑ (c) (5,2),(7,2),(9,5)

❑ (d) (3,4),(4,3),(6,4)

❑ *[GATE 2005]*

<u>A</u>	C
2	4
3	4
4	3
5	2
7	2
9	5
6	4

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GATE Questions



- ☐ The table has two attributes A and C where A is the primary key and C is a foreign key referencing A with on-delete cascade. The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple $(2, 4)$ is deleted is:

- ☐ (a) $(3,4),(6,4)$
☐ (b) $(5,2),(7,2)$
☒ (c) $(5,2),(7,2),(9,5)$
☐ (d) $(3,4),(4,3),(6,4)$

☐ [GATE 2005]

<u>A</u>	C
2	4
3	4
4	3
5	2
7	2
9	5
6	4

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GATE Questions



- ☐ Q. In table $T1$, P is the primary key, Q is the foreign key referencing R in table $T2$ with on-delete cascade. In table $T2$, R is the primary key and S is the foreign key referencing P in the table $T1$ with on-delete cascade. In order to delete record $(3, 8)$ from table $T1$, numbers of additional record that need to be deleted from table $T1$ is

☐ (a) 0

☐ (b) 1

☐ (c) 2

☐ (d) 3

☐ [GATE 2017]

Table T1

<u>P</u>	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

Table T2

<u>R</u>	S
2	2
8	3
3	2
9	7
5	7
7	2

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GATE Questions



- ☐ Q. In table T1, P is the primary key, Q is the foreign key referencing R in table T2 with on-delete cascade. In table T2, R is the primary key and S is the foreign key referencing P in the table T1 with on-delete cascade. In order to delete record (3, 8) from table T1, numbers of additional record that need to be deleted from table T1 is

☐ (a) 0

☐ (b) 1

☒ (c) 2

☐ (d) 3

☐ [GATE 2017]

Table T1

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

Table T2

R	S
2	2
8	3
3	2
9	7
5	7
7	2

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GATE Questions



- ☐ In table T1, P is the primary key, Q is the foreign key referencing R in table T2 with on-delete cascade and on-update cascade. In table T2, R is the primary key and S is the foreign key referencing P in the table T1 with on-delete set NULL and on-update cascade. In order to delete record (3, 8) from table T1, numbers of additional record that need to be deleted from table T1 is

☐ (a) 0

☐ (b) 1

☐ (c) 2

☐ (d) 3

☐ [GATE2017]

Table T1

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

Table T2

R	S
2	2
8	3
3	2
9	7
5	7
7	2

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GATE Questions



- ☐ In table T1, P is the primary key, Q is the foreign key referencing R in table T2 with on-delete cascade and on-update cascade. In table T2, R is the primary key and S is the foreign key referencing P in the table T1 with on-delete set NULL and on-update cascade. In order to delete record (3, 8) from table T1, numbers of additional record ~~that need~~ to be deleted from table T1 is

☒ (a) 0

☐ (b) 1

☐ (c) 2

☐ (d) 3

☐ [GATE2017]

Table T1

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

Table T2

R	S
2	2
8	3
3	2
9	7
5	7
7	2

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GATE Questions



- ☐ Q. Let $R(a,b,c)$ and $S(d,e,f)$ be two relations. 'd' is the foreign key of S that refers to primary key of R. Consider the following four operations on S and R:

(1) Insert into R

(2) Insert into S

(3) Delete from R

(4) Delete from S

Which of these operations may cause violation of referential integrity constraint?

☐ (A) None of (1), (2), (3) or (4) can cause its violation

☐ (B) All of (1), (2), (3) and (4) can cause its violation

☐ (C) Both (1) and (4) can cause its violation

☐ (D) Both (2) and (3) can cause its violation

☐ [GATE 1997]

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GATE Questions



❑ *Q. Let $R(a,b,c)$ and $S(d,e,f)$ be two relations. 'd' is the foreign key of S that refers to primary key of R . Consider the following four operations on S and R :*

- (1) Insert into R*
- (2) Insert into S*
- (3) Delete from R*
- (4) Delete from S*

Which of these operations may cause violation of referential integrity constraint?

- (A) None of (1), (2), (3) or (4) can cause its violation*
- (B) All of (1), (2), (3) and (4) can cause its violation*
- (C) Both (1) and (4) can cause its violation*
- (D) Both (2) and (3) can cause its violation*

❑ *[GATE 1997]*

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GATE Questions



❑ *Q. An ER model of a database consists of entity types A and B . These are connected by a relationship R which does not have its own attribute. Under which of the following conditions, can the relational table for R be merged with that of A ?*

- (a) Relation R is one-to-many and the participation of A in R is total*
- (b) Relation R is one-to-many and the participation of A in R is partial*
- (c) Relation R is many-to-one and the participation of A in R is total*
- (d) Relation R is many-to-one and the participation of A in R is partial*

❑ *[GATE 2017]*

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GATE Questions



❑ *Q. An ER model of a database consists of entity types A and B . These are connected by a relationship R which does not have its own attribute. Under which of the following conditions, can the relational table for R be merged with that of A ?*

(a) Relation R is one-to-many and the participation of A in R is total

(b) Relation R is one-to-many and the participation of A in R is partial

(c) Relation R is many-to-one and the participation of A in R is total

(d) Relation R is many-to-one and the participation of A in R is partial

❑ *[GATE 2017]*

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